

WindOps Copilot - Turbine Report

Turbine ID: WTG-001 | Correlation ID: corr-7f9e8fd67c59
Report Generated: 2026-06-26T14:26:55.030576

TURBINE HEALTH STATUS

65.0% Health

Live Sensor Telemetry

Sensor Parameter	Value	Unit
Wind Speed	14.50	m/s
Rotor Speed	18.00	RPM
Blade Temperature	29.00	°C
Generator Temperature	61.50	°C
Vibration	0.29	mm/s
Power Output	2.80	MW
Operational Status	active	

Active Fault Alerts

Parameter	Value	Threshold	Severity
Vibration	0.29	0.25	CRITICAL

Diagnostic Root Cause Analysis

Diagnosed Cause:	Rotor Imbalance or Blade Misalignment
Confidence Level:	75.0%
Technical Analysis:	Vibrations are elevated at 0.29 mm/s without critical heat signatures. This typical profile points to aerodynamic asymmetry, blade icing, or rotor coupling imbalances.

Operational Action Recommendations

Recommended Action	Description	Risk	Production Loss	Confidence
Emergency Controlled Shutdown	Trigger pitch feathers and rotor brake immediately due to critical vibration threshold breach (>0.25 mm/s). Protect structural integrity of the asset.	LOW	severe	100%

De-rate / Curtail Power Output	Reduce generator speed. (NOT RECOMMENDED: safety thresholds are violated, structural damage is likely if kept spinning).	HIGH	moderate	10%
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Note: This report is generated by an autonomous multi-agent system. final actions must be reviewed and authorized by a human operator.